

Team Members: Nathan Grilliot

Milestones

1. Backend can retrieve metadata for media

When this milestone is achieved, there will be functionality to manually run a function to fetch metadata, and the metadata will be automatically loaded into the database and stored in a structure to allow future retrieval.

2. Backend can automatically sync media history from an external source

Meeting this milestone will allow an account for an external source (Spotify, Steam, etc.) to be manually created, and its data to be automatically imported into the database.

3. Backend can create recommendation based on media history

When this milestone is achieved, a query can be manually written to get the recommendation for the most suitable next media.

4. User can input media manually

Achieving this milestone will require creating a simple UI with an input field and a dropdown menu to create a new media entry.

5. User can connect accounts to sync automatically

To meet this milestone, buttons will be created in the UI to allow OAuth connections to external accounts to automatically sync media history.

6. User can get recommendations automatically.

This milestone is the result of the product, bringing together the user's media history and other media in the database to recommend the next candidate for the user to consume.

7. Queries can be written to create and mutate all nodes in database.

Meeting this milestone will require resolvers to be implemented to all GraphQL queries successfully mutate, create, or delete the correct data.

Table 1: Timeline

Milestone description	Category	Assigned to	Effort (hours)	Progress	Start	End
Backend			145		8/25/2025	1/7/2026
Database			17		8/25/2025	9/24/2025
Design schema for storing media history with fields for books, movies, TV shows, and games.	Goal	Nathan Grilloit	4	40%	8/30/2025	9/7/2025
Create functions to insert information into the database, including handling duplicates and conflicts.	Goal	Nathan Grilloit	6	50%	9/4/2025	9/16/2025
Develop query functions to retrieve media history by type and time range.	Goal	Nathan Grilloit	3	0%	9/9/2025	9/15/2025
Implement mutation functions for database entries.	Goal	Nathan Grilloit	2	10%	9/14/2025	9/18/2025
Document database functions with examples for future developers.	Goal	Nathan Grilloit	2	0%	9/19/2025	9/23/2025
Backend can create recommendation based on media history	Milestone	Nathan Grilloit		20%	9/24/2025	9/24/2025
App Integrations			22		9/29/2025	10/17/2025
Research applications that provide APIs to retrieve user data across books, movies, TV, and games.	Goal	Nathan Grilloit	3	50%	10/1/2025	10/7/2025
Implement API clients to connect to at least three external services.	Goal	Nathan Grilloit	4	0%	10/3/2025	10/11/2025
Handle authentication flows for third-party APIs securely.	Goal	Nathan Grilloit	6	0%	10/5/2025	10/17/2025
Test data retrieval from each integrated service with mock user accounts.	Goal	Nathan Grilloit	5	0%	10/7/2025	10/17/2025
Document API integration steps and usage patterns.	Goal	Nathan Grilloit	4	0%	10/9/2025	10/17/2025
Backend can automatically sync media history from an external source	Milestone	Nathan Grilloit		10%	10/11/2025	10/17/2025
GraphQL API			34		9/15/2025	1/7/2026
Update resolvers to ensure data is retrieved correctly from the database.	Goal	Nathan Grilloit	4	50%	9/27/2025	10/5/2025
Design GraphQL queries and mutations for CRUD operations on media history.	Goal	Nathan Grilloit	8	0%	10/21/2025	11/6/2025
Implement middleware for logging and authentication in the GraphQL layer.	Goal	Nathan Grilloit	6	0%	11/8/2025	11/20/2025
Test GraphQL API endpoints with Postman and automated test scripts.	Goal	Nathan Grilloit	10	0%	12/8/2025	12/28/2025
Queries can be written to create and mutate all nodes in database.	Milestone	Nathan Grilloit	6	0%	12/26/2025	1/7/2026
Metadata			20		9/22/2025	10/17/2025
Research methods of accessing media-related metadata for books, movies, TV shows, and games.	Goal	Nathan Grilloit	4	75%	9/25/2025	10/3/2025
Implement a metadata fetching pipeline to add metadata to inputted media.	Goal	Nathan Grilloit	8	20%	9/28/2025	10/14/2025
Validate metadata consistency across different providers.	Goal	Nathan Grilloit	8	0%	10/1/2025	10/17/2025
Backend can retrieve metadata for media	Milestone	Nathan Grilloit		0%	8/25/2025	10/17/2025
Frontend			52			11/28/2025
CRUD Operations			32		10/5/2025	11/10/2025
Develop basic user flows to create, read, update, and delete a user's media history.	Goal	Nathan Grilloit	10	0%	10/9/2025	10/29/2025
Design UI components for displaying media history entries.	Goal	Nathan Grilloit	4	0%	10/13/2025	10/21/2025
Implement input forms with validation for adding new media items.	Goal	Nathan Grilloit	8	0%	10/17/2025	11/2/2025
Test CRUD operations with dummy data and API responses.	Goal	Nathan Grilloit	10	0%	10/21/2025	11/10/2025
User can input media manually	Milestone	Nathan Grilloit		0%	10/25/2025	10/25/2025
OAuth Connections			20		11/3/2025	11/28/2025
Create a way to connect to external platforms with OAuth.	Goal	Nathan Grilloit	6	0%	11/6/2025	11/18/2025
Implement a secure token storage mechanism on the client.	Goal	Nathan Grilloit	6	0%	11/9/2025	11/21/2025
Validate OAuth flows by testing with multiple external accounts.	Goal	Nathan Grilloit	8	0%	11/12/2025	11/28/2025
User can connect accounts to sync automatically.	Milestone	Nathan Grilloit		0%	11/15/2025	11/15/2025
User can get recommendations automatically.	Milestone	Nathan Grilloit		0%	8/25/2025	1/7/2026

Table 2: Effort Matrix

As the only team member all tasks are assigned to me. Effort for each task is listed in the table above.